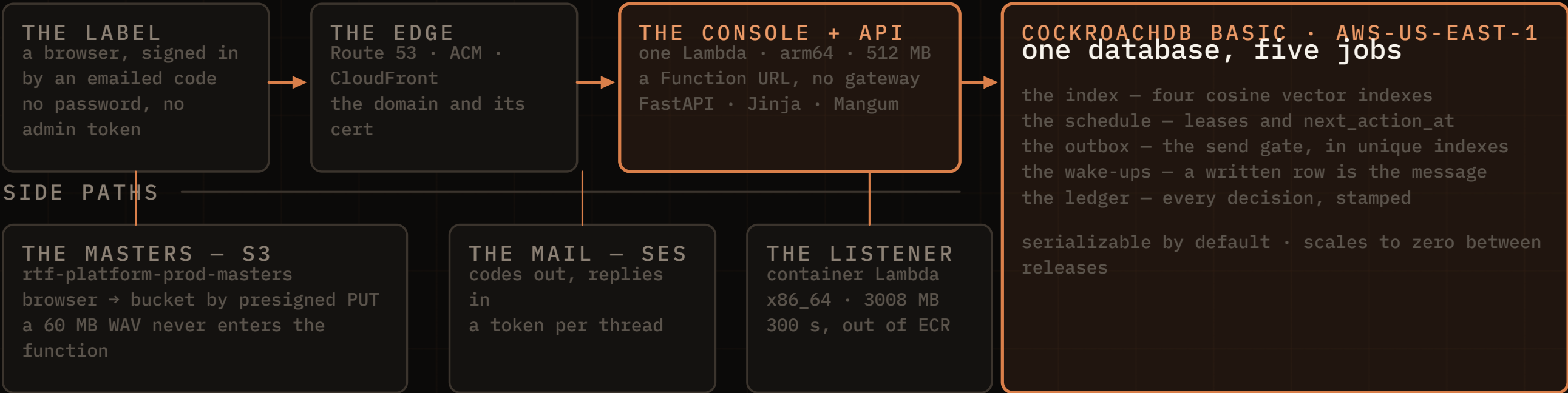


An agentic OS for a record label, drawn as what is actually deployed.

Two Lambdas, one bucket, one mail identity and one database. Every box below is provisioned and running; anything written but not switched on is on **page 5**, with the command that proves it. Sizes and names are read from the applied Terraform state, not from the plan.

REQUEST PATH



THE FLEET – NO ORCHESTRATOR

17 AGENTS, AND NOT ONE OF THEM CALLS ANOTHER

- embed_document
- map_source
- find_counterparties
- profile_party
- embed_party
- distil_lesson

- analyse_recording
- index_stations
- index_streams
- enrich_genre
- index_podcasts
- refresh_stream

- harvest_contacts
- harvest_feed_contacts
- profile_creator
- refresh_audience
- profile_audience

each finishes by writing a row; that row is what wakes the next one

THE GATE
a person approves
every send
the one irreversible
act

A MUSIC DIRECTOR
a named human being,
who can reply or ask us
to stop

the reply lands back in the index
and reranks the next shortlist

nothing here bills while idle

The memory, and the four indexes the fleet decides with.

Every filter that must always hold sits **inside the index prefix** rather than in a WHERE clause applied afterwards – so a search cannot return a row belonging to another label, a stale embedding model, or somebody who cannot be contacted.

THE FOUR VECTOR INDEXES · COSINE · FILTERS IN THE PREFIX

party_shortlist

ON PARTY

who, out of everyone indexed, should hear this particular record

tenant_id

embedding_model

party_class

contact_state

the embedding

fact_semantic

ON PARTY_FACT

what we have concluded about them – superseded facts cannot surface

tenant_id

model

status

the embedding

chunk_semantic

ON PARTY_CHUNK

the source text a conclusion was drawn from

tenant_id

model

the embedding

lesson_semantic

ON LESSON

what we learned that reranks the next shortlist

tenant_id

model

scope_kind

the embedding

Equality on the prefix is both the correctness rule and the fast path. A post-filter is neither: it over-fetches to compensate, and any caller writing a query can forget it.

AND THE CONSTRAINTS THAT MAKE SENDING SAFE

one_open_thread_per_counterparty · outbox UNIQUE (message_id) · message UNIQUE (tenant_id, idempotency_key)

54 TABLES, BY WHAT THEY HOLD

- THE INDEX**

party · party_fact · party_chunk · party_role · party_relation · party_identifier · party_alias · contact_route · source_manifest
- THE ROSTER**

artist · artist_profile · artist_fact · artist_document · recording · release · work · track · sound_usage · audience_profile
- OUTREACH**

campaign · thread · message · outbox · lead · inbound_unmatched · presence
- MEMORY**

lesson · decision · fact_basis · agent_run · agent_manifest · suggestion
- MONEY**

account · tenant_budget · artist_budget · recording_budget · budget_raise · autonomy
- ACCESS**

tenant · otp_challenge · mail_account · listen_token · listen_event

These are index structure, **not predicates** somebody has to remember.

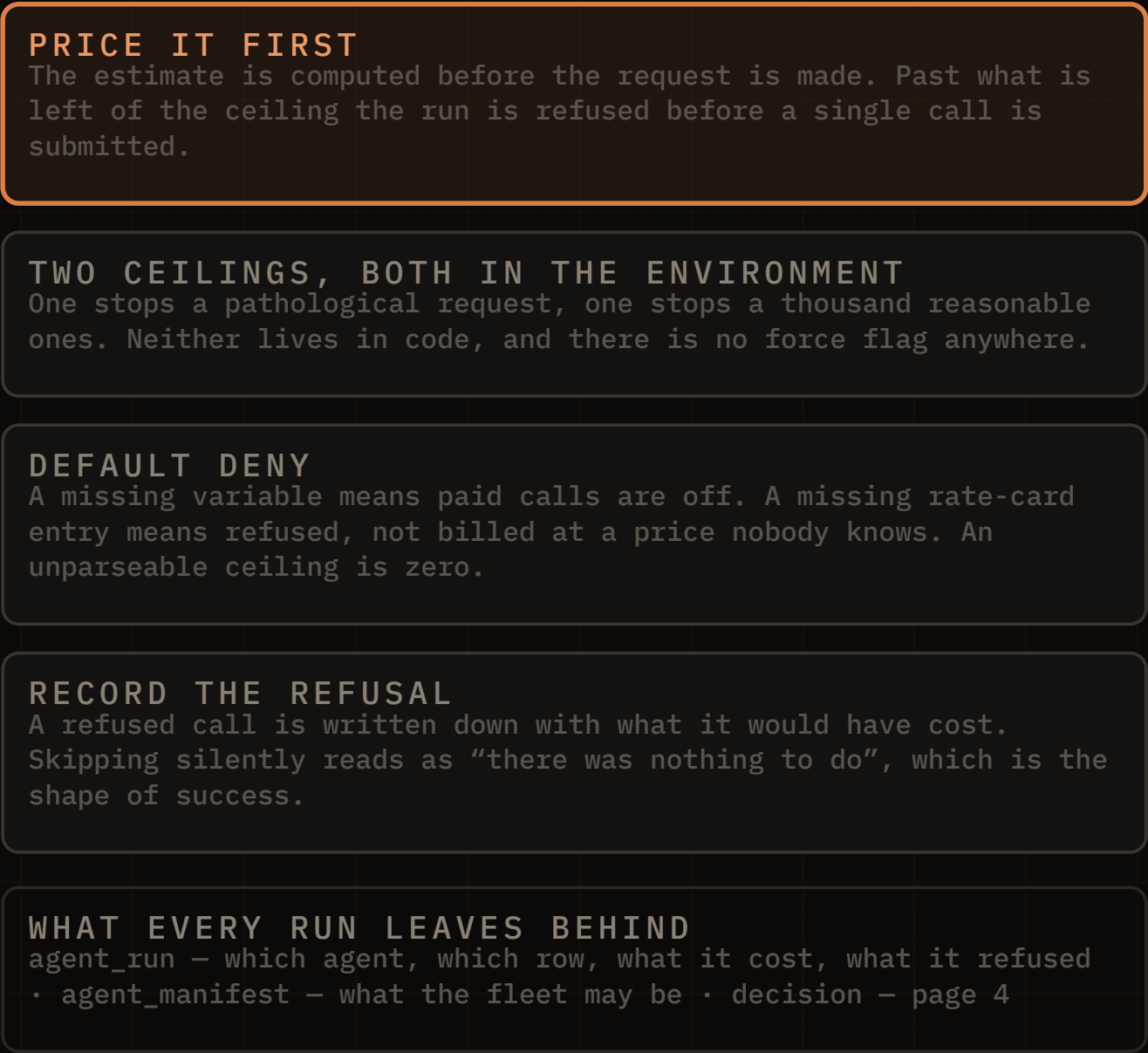
The database is the scheduler, the queue and the message bus.

No agent calls another. An agent finishes by writing a row with a `next_action_at`, and that row is the message. Take the database away and the system does not slow down – it loses its control flow entirely.

HOW WORK IS CLAIMED, AND SURVIVES A CRASH



BEFORE ANY PAID CALL LEAVES THE BUILDING



Why did the agent decide that, then?

The memory is alive: lessons accumulate, embeddings land, contact states move. By morning the world a decision was made in is gone. The ledger stores the exact logical timestamp, and the same query pointed at that timestamp returns the index as it stood – not a copy of it.

THE LEDGER ROW

DECISION

tenant_id the label it belongs to
kind what sort of decision
stage proposed, or applied
at_hlc the hybrid logical clock, as a decimal
at_wall the same instant, in human time
subject_kind what it was about
actor which agent, or which person
summary one sentence, for a human
inputs the numbers it was looking at

at_hlc is the whole trick. It is not a note about when something happened – it is the value the database itself will accept as “then”.

THE SEVEN KINDS

shortlist	somebody was selected out of the index
send	a message left the building
budget_increase	a spend cap was raised
plan_change	the tier moved
opt_out	somebody asked us to stop – terminal
suppress	we decided NOT to contact somebody
lesson	something learned that reranks later

ANSWERING “WHY DID YOU DO THAT?” THE NEXT MORNING

LAST NIGHT – IT DECIDED

The fleet ranked the index, opened a thread, and asked to put more behind one record. A person approved it, and the ledger row stored the instant.

BY MORNING, THE WORLD HAS MOVED

Metrics overwritten, contact states changed, lessons landed and reordered the ranking. Asking now returns today’s answer wearing last night’s clothes.

THE SAME QUERY, POINTED AT THAT INSTANT

```
SELECT ... FROM party@party_shortlist AS OF SYSTEM TIME '<at_hlc>' WHERE
tenant_id = $1 AND embedding_model = $2 AND party_class = $3 AND
contact_state = $4 ORDER BY profile_embedding <=> $5 LIMIT 20
```

WHAT COMES BACK IS THE INDEX ITSELF

Same index, same embeddings, same lesson rows, as they stood. No schema was written for it and the write path pays nothing. Not a copy of them – them.

AND THEN IT CHECKS ITSELF

The valuation is re-run against those old numbers and compared with what was acted on. Agreeing is the point; disagreeing is the finding.

PAST THE RETENTION WINDOW, IT REFUSES

Beyond gc.ttlseconds the versions are genuinely collected, and the function raises rather than quietly dropping the clause and answering anyway.

The parts that are written and deliberately not switched on.

A claim a judge can falsify in one command is worse than no claim. Each item below is code that exists, is tested, and has been left off for a reason – with the command that returns nothing, printed beside it.

WRITTEN, TESTED, AND NOT SWITCHED ON

THE CHANGEFEED

The statement, the consumer, the Lambda handler and a --verify path all exist. Nothing has been created on this cluster, so the fleet is woken by polling next_action_at rather than by a feed.

proven, not created

WHAT A JUDGE RUNS

SHOW CHANGEFEED JOBS → zero rows

REGIONAL BY ROW DOMICILING

Migration 024 and the multi-region Terraform are written and validate. They are not applied: adding a region to a Basic cluster is irreversible, and converting the cluster holding the system of record would be a one-way door taken for a demo. It was shown on a throwaway cluster instead, by a script that refuses to run against a single-region one.

proven, not deployed

WHAT A JUDGE RUNS

SHOW REGIONS → aws-us-east-1, and nothing else

BEDROCK

Both the on-demand and the batch path are implemented, and neither can execute on this account: on-demand quota is zero and non-adjustable, batch inference is entitlement-gated. Embeddings run through OpenAI instead.

written, cannot run

WHAT A JUDGE RUNS

the quota console → 0, non-adjustable

AND HOW TO CHECK EVERYTHING ELSE

```
/          every counter across the top is a query run on page load
/#tune     move the dial and the shortlist reranks, live
/#plan     the EXPLAIN output: the filters are inside the index
/#send     the send gate, and why a double send cannot happen
/#verdict  where Postgres would have done fine, feature by feature
/manual    the operator manual, written for label staff
```

and in the repository:
test_vector_plans.py asserts the plan, not a comment
test_sender.py names the double-send case it forbids
the sponsor audit argues this project against itself